

Data Analysis for Cognitive Neuroscience 2026/27



**Universitat
Pompeu Fabra**
Barcelona

Klaus Wimmer

kwimmer@crm.cat

Alexandre Hyafil

ahyafil@crm.cat

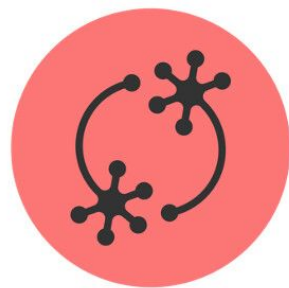


Inspirations

Albert Compte

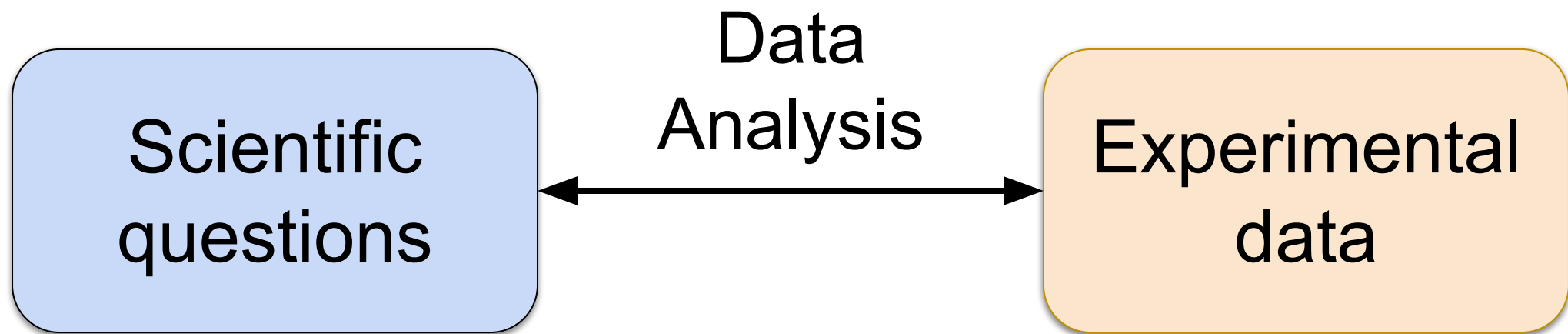


www.bambschool.org



neuromatch
academy

<https://academy.neuromatch.io/>



What is the aim of this course?

- Data analysis is fun!
- Data analysis is creative!
- Data analysis has constraints:
 - rigor
 - some structure
 - data checking / cleaning
 - preprocessing
 - exploratory data analysis
 - statistical inference
 - model-based analyses
 - presentation of results
- Learn specific general-purpose techniques

How will we achieve that??

- Working on **real data** as examples
- Using **Python** as a programming language
- We will learn: statistical inference, linear models, logistic regression, dimensionality reduction, cluster analysis (exploratory), drift-diffusion model (a model for perceptual decision-making)

Classroom / Assignments dynamics

- Before each class: we'll send the theoretical material (mostly videos) to be watched *prior* to the class
- During the class:
 - we'll make a short recap, present real examples and address your questions
 - Then we'll start doing the tutorial in class
- After the class: You will have to complete it and send it by *Wednesday* night.

Feel free to work your tutorials in pairs, during and after class, but each student needs to upload their assignment

Why learn coding?

- Because it opens a world of opportunities for data analysis, programming experiments, etc., that are simply not possible with button-clicking solutions (SPSS, Presentation,...)
- Because you reach a deeper understanding of the analysis
- Because this has become a very valuable asset in the industry (specifically Python for data scientists)

Coding in the era of ChatGPT

- AI is your friend, but the type of friend you can never really trust
- Which is why you need to build the ability to evaluate ChatGPT's output and dialogue with it
- Speeds up learning and can allow you to focus more on the big picture
- But never trust it!

Coding in the era of AI coding assistants

A sparring partner, not an oracle



The oracle

PASSIVE USE

Paste the question, copy the answer, move on.

Unrestricted chatbot access before an exam lowered students' performance in the real exam. (Bastani et al., PNAS 2025)



The sparring partner

ACTIVE DIALOGUE

Think first, then ask. Question the answer, test it, argue back.

The same model set up as a hint-giving tutor did not hurt performance, and working through ideas first led to stronger engagement. (Bastani et al., 2025; Kosmyna et al., 2025 preprint)

Four habits of an active user

Friend, yes. But the kind you never fully trust.



1

Think first

Sketch your approach before you prompt.



2

Challenge

Ask for assumptions, edge cases and alternatives.



3

Verify

Run it, test it, and check it against the data and the docs.



4

Record

Note what you accepted, what you rejected, and why.

The skill that matters: question it, spot the flaws, reject weak answers.

Source: Schmitt, "Use AI as a sparring partner, not an oracle", Nature 657, 575 (2026)

Schedule

Date	Topic	Discuss in class/Submit	Get	Teacher
Oct 2	Introduction to Python		Assignment 0	Alex & Klaus
Oct 9	Descriptive statistics	Assignment 0	Assignment 1	Klaus
Oct 16	Linear models	Assignment 1	Assignment 2	Klaus
Oct 23	Psychometric curves	Assignment 2	Assignment 3	Klaus
Oct 30	Logistic regression	Assignment 3	Assignment 4	Alex
Nov 6	Statistical inference	Assignment 4	Assignment 5	Alex
Nov 13	Modeling Recap	Assignment 5	Assignment 6	Alex
Nov 20	Drift-diffusion model (DDM)	Assignment 6	Assignment 7	Alex
Nov 27	Dimensionality reduction / Decoding	Assignment 7	Assignment 8	Klaus
Dec 4	Journal Club	Assignment 8		Alex & Klaus
Dec 11	Exam			Alex & Klaus



General Course Information



Tauler d'avisos



Schedule, program and evaluation



Course Homepage on Github

All information about the course will be on github. By clicking on the URL you will get to a readme file that contains the information about how to get started in Python and several Python tutorials. In the future, we will post the assignments there as well.

before January 12



Basic Python programming quiz

Marca com a feta

Before the course starts on January 12, do the quick quiz about your Python programming skills, so you know if you need to do an extra push before the class starts (**see the link to our github page above for links to tutorials etc.**). We will not consider this quiz for the final evaluation of the course. It is just for you to see where you stand and what concepts you should be familiar with in order to benefit from the class. If you

https://github.com/wimmerlab/MBC_data_analysis

MBC-DataAnalysis

This is the repository for the materials of the Data Analysis class at the UPF Masters for Brain & Cognition taught by Klaus Wimmer & Alex Hyafil, largely using material from Albert Compte.

Coding in Python

Why should a cognitive psychologist / neuroscientist learn how to code? See the arguments in these different sources:

- "The next generation of neuroscientists needs to learn how to code" [article from Neuron journal](#)
- "Why it's important to learn to code in contemporary biology?" [article from Wire](#)
- "Why every (psychology) student should learn to code" [blog post](#)

You can brush up your Python skills by following one (or some) of these suggested Python tutorials:

- [Introduction to Python](#) from the Department of Cognitive Science of UC San Diego
- [Python for Everybody Specialization](#) (free Coursera course starts Dec 20)
- [LearnPython.org](#)
- [Datacamp - Intro to Python for Data Science](#)
- Python training at the [Neuromatch Academy: Day 1 and Day 2](#), tutorials with videos
- Introductory material from the "[Advanced Scientific Programming in Python](#)" summer school
- If your master's project relies heavily on programming or you continue along with a PhD, we strongly recommend reading quietly through Patrick Mineault's wonderful [Good Research Code Handbook](#)

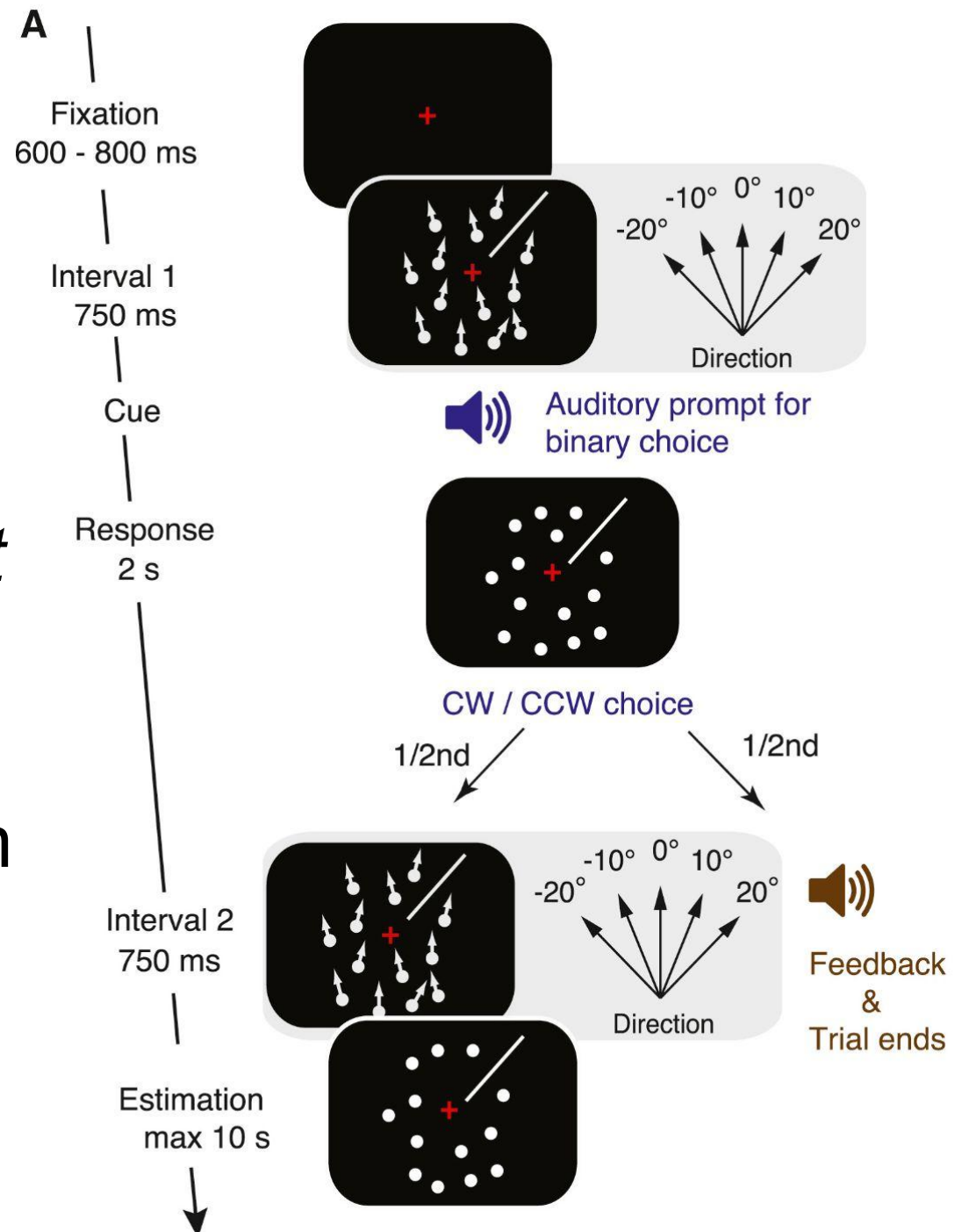
Evaluation

The grade of the course will be computed based on:

- 1) **Completed assignments.** Students will submit the code generated for each assignment and their interpretation of the results. Within-deadline submission and quality of the assignments will count towards **40% of the grade.**
- 2) **Final exam.** At the end of the trimester there will be an exam, including both a Python Notebook assignment (no AI support allowed!) and questions about the interpretation of data analysis results. It will count towards **60% of the grade.**

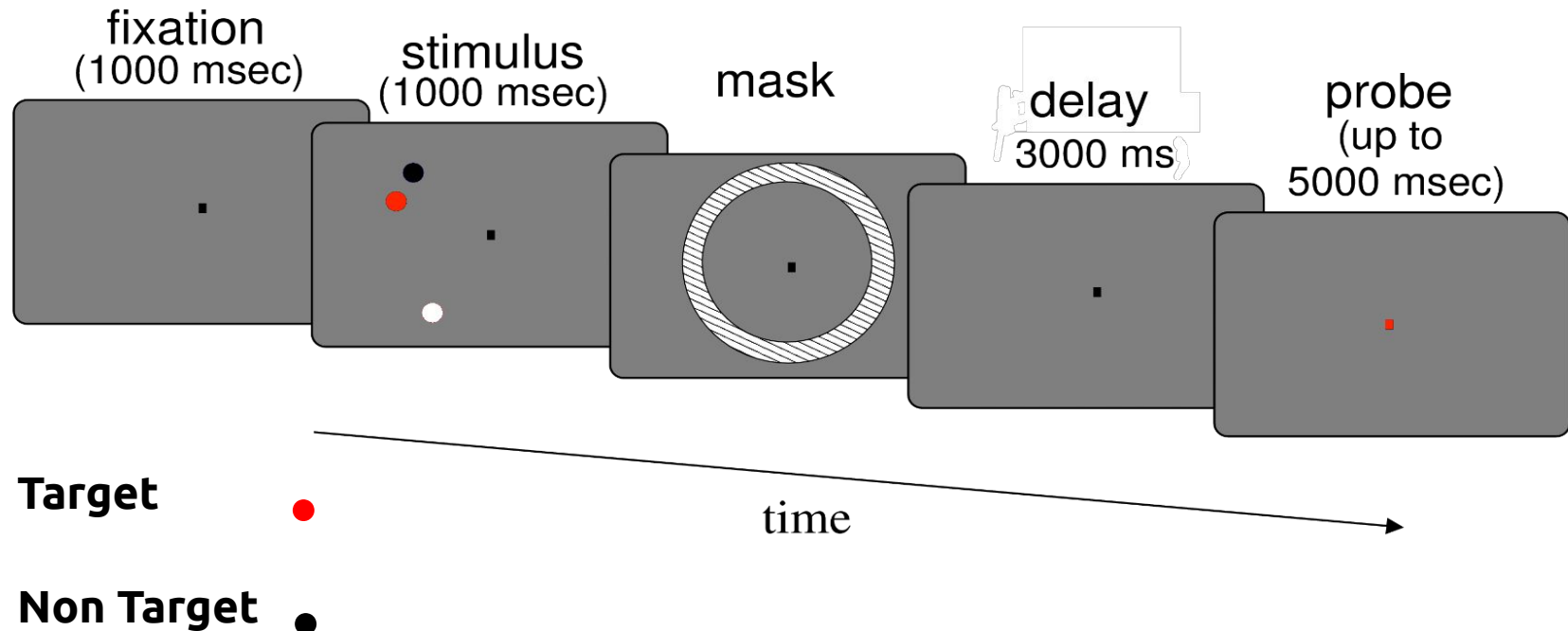
Dataset 1: behavioral data

- Talluri et al. *Current Biol* 2018
- combined discrimination/estimation task
- 14 subjects
- confirmation bias



Dataset 2: psychophysics data

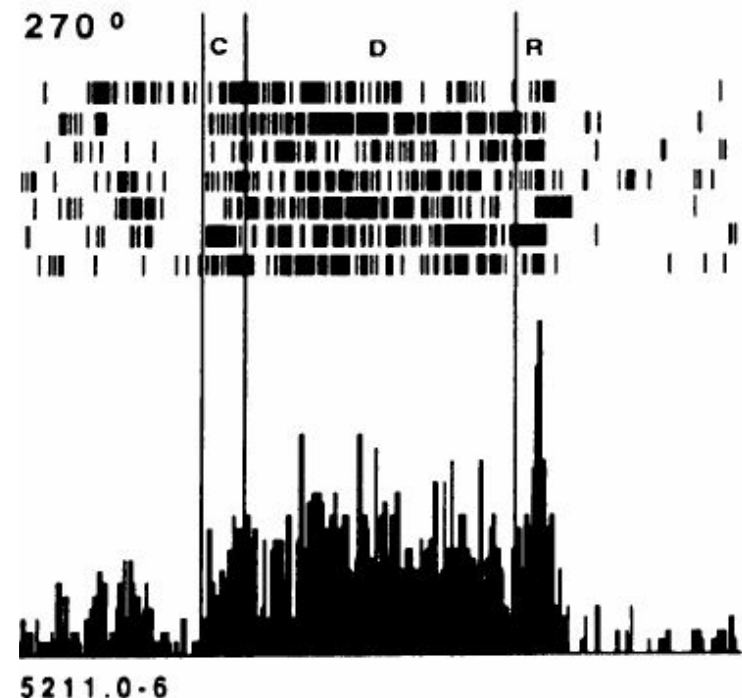
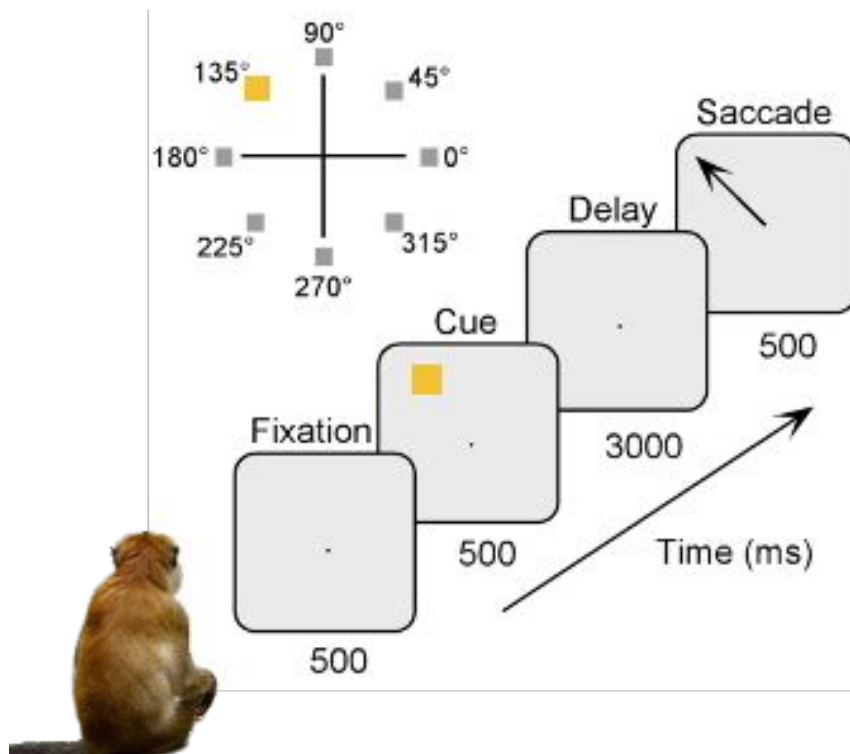
- Almeida et al. *J Neurophysiol* 2015
- working Memory (WM) experiments: 2AFC
- 9 subjects



Dataset 3: working memory

[Constantinidis et al. J. Neuroscience 2001](#)

>200 neurons recorded from prefrontal cortex



Assignment 0

Things to learn about Python

- Basic maths
- Arrays
- Indexing (most analyses rely on computing something or one or several subsets of the dataset)
- Plotting
- Functions and scripts
- Dataframes (a fantastic tool to handle datasets)

Assignment 1

- Descriptive statistics
 - Measures of centrality
 - Measures of dispersion
- Statistical inference
 - t -tests
 - Permutation tests